AI Lab Project

Project Description for Students

You are required to design and develop an **AI-based system** that solves a real-world problem using the concepts studied in Artificial Intelligence Lab. Your project may belong to any domain such as education, health, transport, games, environment, agriculture, finance, campus management, recommendation systems, or smart decision support.

Although topics may be different, every project must show how AI can be used to make decisions, search for solutions, classify data, cluster information, optimise results, or reason logically.

Minimum Project Requirements

Each group must include **at least two AI components** from the following:

1. **Search / Path-Finding / Problem Solving**
   Example algorithms: BFS, DFS, Depth-Limited Search, IDS, Best First Search, A*.
2. **Machine Learning / Data-Based AI**
   Example algorithms: KNN, K-Means Clustering, Naive Bayes Classification.
3. **Optimisation / Game AI**
   Example algorithms: Genetic Algorithm, Minimax, Alpha-Beta Pruning.
4. **Logic-Based Reasoning**
   Example tools/concepts: Prolog, predicates, clauses, First Order Logic rules.

The system must be implemented practically using **Python and/or Prolog**, depending on the selected topic.

Suggested Project Ideas

Students can choose any topic, but here are examples:

Domain	Possible Project Idea	Suitable AI Techniques
Campus	Smart classroom allocation system	Search + Genetic Algorithm
Transport	Shortest route finder for university buses	A* + BFS
Games	Tic-Tac-Toe / Connect Four AI player	Minimax + Alpha-Beta
Healthcare	Disease prediction from symptoms	Naive Bayes + KNN
Education	Student performance prediction	KNN + Naive Bayes
Environment	Waste classification system	KNN + Naive Bayes
Agriculture	Crop recommendation system	KNN + Rule-based logic
Security	Spam message detection	Naive Bayes

Shopping	Customer grouping system	K-Means clustering
Career	Job recommendation system	KNN + rule-based reasoning
Emergency	Rescue path planning system	A* + Best First Search
Scheduling	Exam timetable optimisation	Genetic Algorithm
Logic AI	Expert system for laptop/mobile recommendation	Prolog + FOL rules
Robotics	Grid-based robot path planning	BFS + A*
Sports	Team/player performance grouping	K-Means + classification

Students should not simply copy these titles. They may modify them or propose their own.

Project Deliverables — 40 Marks

1. Working Implementation — 14 Marks

The project must include:

- Working code
- Proper input/output
- AI algorithm implementation
- Comments and modular structure
- Demonstration of results
- Error handling where needed

2. AI Technique Application — 8 Marks

Students must clearly show:

- Why the selected AI technique is suitable
- How the algorithm works in their project
- Step-by-step logic or flow
- Comparison of outputs, if applicable
- Correct interpretation of results

3. Testing and Evaluation — 4 Marks

Depending on the project type, students should include:

- Accuracy, precision, recall, confusion matrix, or error rate for ML projects
- Path cost, number of nodes expanded, execution time for search projects

- Fitness value, convergence, or best solution for genetic algorithm projects
- Win/loss/draw behaviour for game AI projects
- Rule validation for Prolog/expert system projects

4. Report — 6 Marks

The report should contain:

- Abstract
- Introduction
- Problem statement
- Literature/background
- Methodology
- Algorithm explanation
- Screenshots/results
- Evaluation
- Ethical considerations
- Limitations
- Conclusion
- References

7. Presentation and Viva — 4 Marks

Students must present:

- Problem overview
- AI method used
- System demo
- Results
- Challenges faced
- Individual contribution
- Viva answers

Common Rubric for All Projects — 40 Marks

Criteria	Marks	Excellent	Good	Satisfactory	Poor
Problem Selection and Proposal	4	Clear, meaningful real-world problem with	Clear problem with reasonable justification	Basic problem, limited justification	Unclear or copied topic

		strong justification			
AI Relevance	4	AI technique is highly suitable and well justified	Technique is suitable but justification is brief	Some AI relevance, weak explanation	Technique does not match problem
Algorithm Implementation	6	Correct, functional, and mostly self-written implementation	Working implementation with minor issues	Partially working implementation	Non-working or copied code
Integration of Two AI Components	4	Two AI components are meaningfully integrated	Two components included but integration is basic	Components are separate or weakly connected	Only one or no AI component
Dataset / Input Handling	3	Clean input/data handling with preprocessing where needed	Input/data handled with minor issues	Basic input/data handling	Poor or missing input handling
Testing and Evaluation	4	Strong evaluation with suitable metrics and interpretation	Evaluation included but limited	Basic testing only	No proper testing
Code Quality	3	Modular, readable, commented, and well-structured	Mostly readable with some comments	Messy but understandable	Disorganised or difficult to run
User Interface / Usability	2	Easy to use, clear outputs, good interaction	Usable with minor issues	Basic command-line output	Confusing or unusable
Report Quality	5	Complete, well-organised, technically sound	Mostly complete with minor gaps	Basic report with missing depth	Poor, incomplete, or unstructured
Ethics and Professional Practice	2	Clearly discusses privacy, bias, misuse, limitations	Mentions ethics briefly	Very limited ethics discussion	No ethics discussion

Presentation and Viva	3	Confident demo and strong answers	Good demo and acceptable answers	Basic presentation	Poor demo or unable to answer

Total: 40 Marks